

Lock-In Amplifier

PCB Design and Working Principle

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1. Introduction

A lock-in amplifier is an instrument used to measure very small AC signals buried in noise. It works by multiplying the input signal with a reference signal of known frequency and phase, followed by low-pass filtering. This process extracts the component of the signal that is synchronous with the reference frequency while rejecting noise at other frequencies.

The implemented circuit consists of the following blocks:

- Input Preamplifier
- Reference Generation (0° and 90° square waves)
- Phase Sensitive Detection using AD630
- Low Pass Filtering
- Amplitude and Phase Calculation

The schematic of the circuit contains two AD630 demodulators producing **in-phase (X)** and **quadrature (Y)** outputs.

2. System Overview and Block Diagram

The system is divided into four functional blocks. The signal flows from the input through the pre-amplifier and phase-sensitive detector, while the reference generator feeds the quadrature clocks from below, and the power supply feeds all blocks.

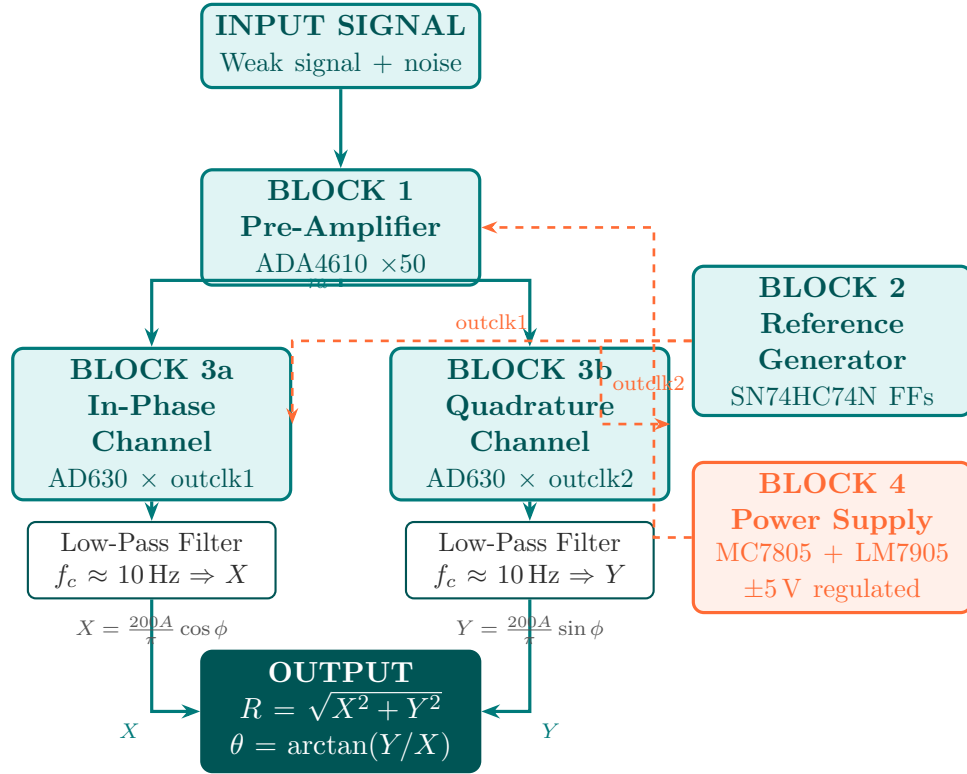


Figure 1: System block diagram of the Lock-In Amplifier. Solid teal arrows show the signal path; dashed orange arrows show reference and power connections.

2.1. Circuit Implementation

The complete schematic of the designed lock-in amplifier is shown in Fig. 2. The circuit is divided into four major functional blocks: the input preamplifier, the reference signal generation block, the phase-sensitive detection stage using the AD630, and the low-pass filter stage for extracting the DC output.

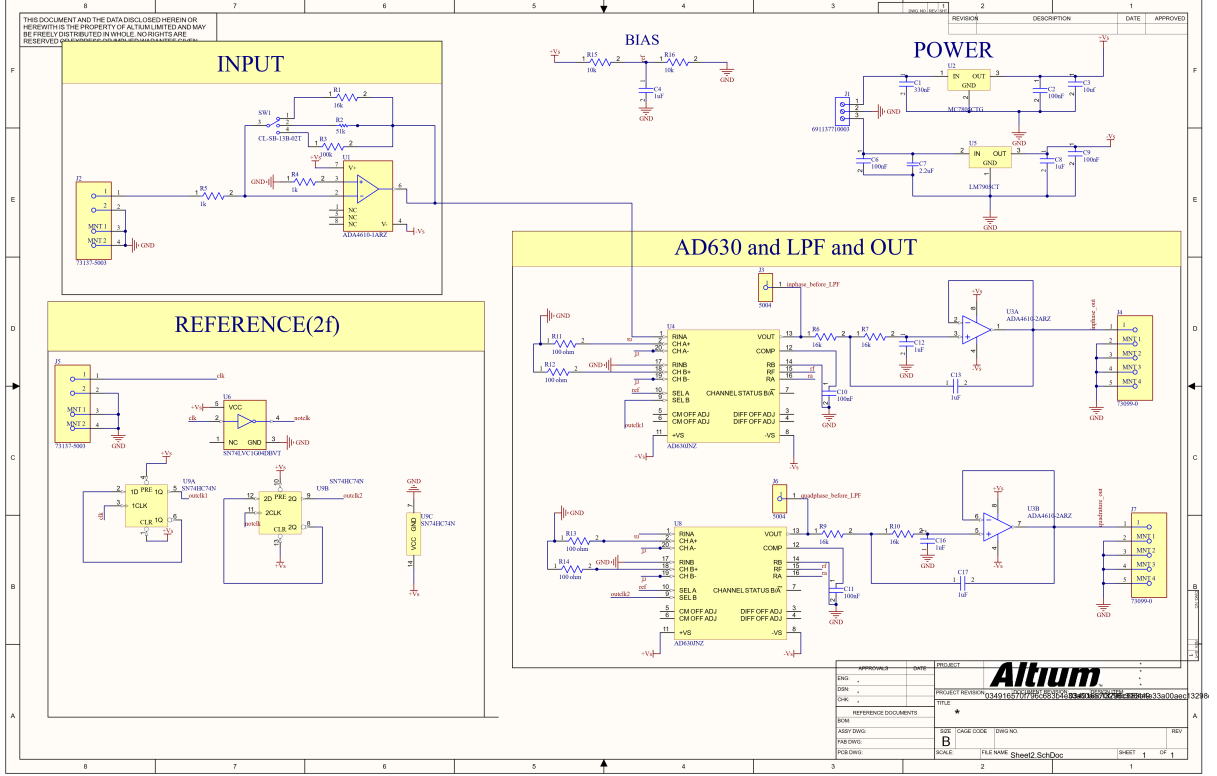


Figure 2: Complete schematic of the dual-phase lock-in amplifier

2.2. Input Preamplifier

The input signal is typically very small and buried in noise. Therefore the first stage of the circuit is a low-noise preamplifier.

Let the input signal be

$$v_{in}(t) = A \sin(\omega t + \phi) \quad (1)$$

After amplification with gain G_{pre} :

$$v_p(t) = G_{pre} A \sin(\omega t + \phi) \quad (2)$$

In the implemented circuit the preamplifier stage inverts the signal. However, the reference signal applied to the AD630 comparator is also inverted. This ensures that the overall polarity of the synchronous detection remains correct.

The amplified signal is then applied to both AD630 demodulators.

2.3. Reference Generation

The reference signal is supplied at frequency $2f$.

A NOT gate followed by two positive-edge-triggered D flip-flops is used to divide the frequency and generate two square waves:

- Reference 1 : frequency f with phase 0°
- Reference 2 : frequency f with phase 90°

Because the comparator input of the AD630 is connected in reverse polarity, the effective reference waveform becomes

$$r(t) = -\text{sgn}(\sin(\omega t)) \quad (3)$$

The Fourier series expansion of this square wave is

$$r(t) = -\frac{4}{\pi} \left(\sin(\omega t) + \frac{1}{3} \sin(3\omega t) + \frac{1}{5} \sin(5\omega t) + \dots \right) \quad (4)$$

After multiplication and low-pass filtering, only the fundamental component contributes to the final DC output.

2.4. Phase Sensitive Detection

The AD630 performs synchronous multiplication of the amplified input signal with the reference square wave.

The input to the demodulator is

$$v_p(t) = G_{pre} A \sin(\omega t + \phi) \quad (5)$$

Multiplying with the reference square wave:

$$v_m(t) = v_p(t) \cdot r(t) \quad (6)$$

Substituting the fundamental component of the square wave:

$$v_m(t) = G_{pre} A \sin(\omega t + \phi) \cdot \left(-\frac{4}{\pi} \sin(\omega t) \right) \quad (7)$$

Using the trigonometric identity

$$\sin a \sin b = \frac{1}{2} [\cos(a - b) - \cos(a + b)] \quad (8)$$

$$v_m(t) = -\frac{4G_{pre}A}{2\pi} [\cos(\phi) - \cos(2\omega t + \phi)] \quad (9)$$

2.5. Low Pass Filtering

The low-pass filter removes the 2ω component.

The remaining DC component becomes

$$X = -\frac{2}{\pi}G_{pre}A \cos(\phi) \quad (10)$$

Similarly, for the quadrature channel (90° shifted reference):

$$Y = -\frac{2}{\pi}G_{pre}A \sin(\phi) \quad (11)$$

2.6. AD630 Gain

The AD630 is configured with gain

$$G_{AD630} = 2 \quad (12)$$

Therefore the final outputs become

$$X = -\frac{4}{\pi}G_{pre}A \cos(\phi) \quad (13)$$

$$Y = -\frac{4}{\pi}G_{pre}A \sin(\phi) \quad (14)$$

The negative sign indicates an inversion introduced by the reference polarity and preamplifier configuration. This does not affect the amplitude calculation, since the magnitude is obtained from

$$R = \sqrt{X^2 + Y^2} \quad (15)$$

which removes the polarity dependence.

2.7. Amplitude and Phase Extraction

From the measured outputs X and Y :

$$R = \sqrt{X^2 + Y^2} \quad (16)$$

Substituting:

$$R = \frac{4}{\pi}G_{pre}A \quad (17)$$

Therefore the true signal amplitude is

$$A = \frac{\pi}{4G_{pre}} R \quad (18)$$

The phase of the signal is

$$\theta = \tan^{-1} \left(\frac{Y}{X} \right) \quad (19)$$

3. Simulation Results

3.1. Circuit Schematic

The Lock-In Amplifier circuit was implemented and simulated in LTspice. The complete simulation schematic is shown in Figure 3, consisting of the input signal source, the AD630 synchronous demodulator, and the second-order active low-pass filter at the output stage.

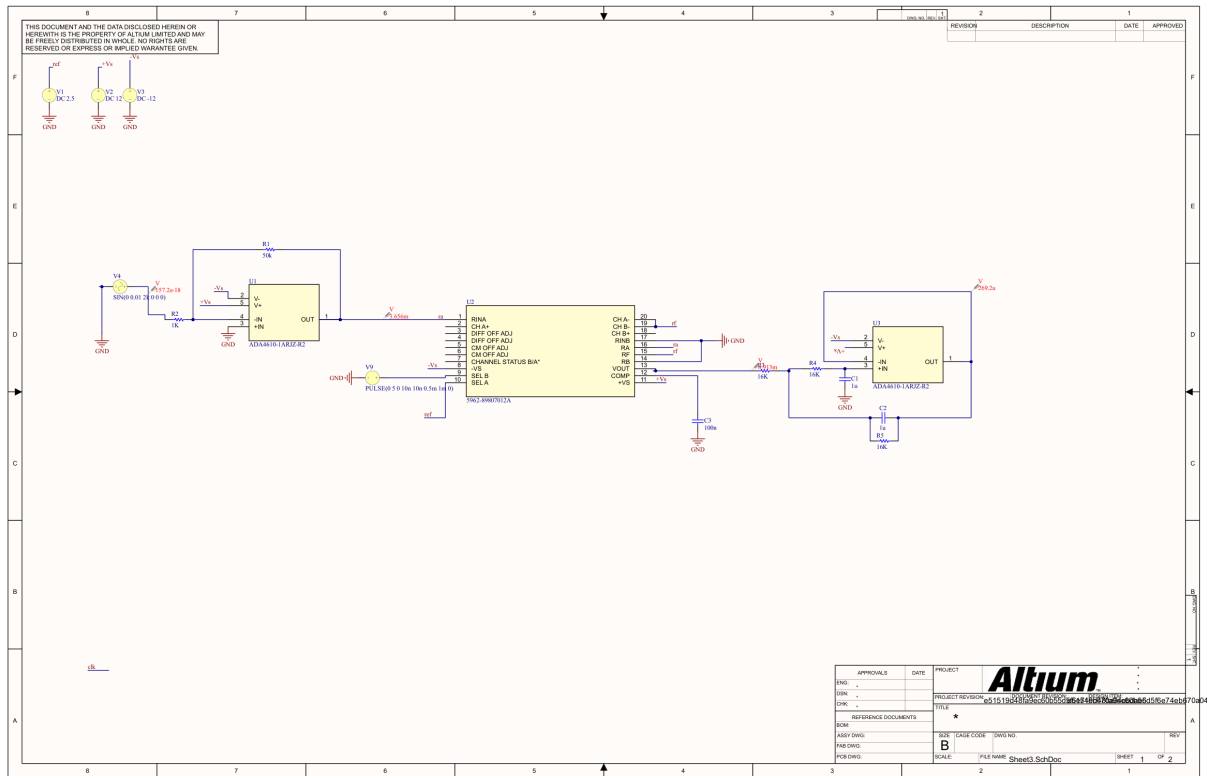


Figure 3: LTspice simulation schematic of the Lock-In Amplifier showing the AD630 demodulator and second-order active low-pass filter.

3.2. Simulation Parameters

Table 1: Key Simulation Parameters

Parameter	Value
Input signal amplitude	10 mV
Reference frequency	1 kHz
Supply voltage	± 12 V
LPF cutoff frequency	≈ 10 Hz
Simulation duration	150 ms

3.3. Output Settling Time

Due to the second-order active low-pass filter at the output stage, the DC output of the LIA does not respond instantaneously. The filter time constant $\tau = RC$ introduces a settling delay, and the output is expected to reach its final steady-state DC value in approximately **200 ms**. Therefore, all simulation results are captured after this settling period to ensure the measured output reflects the true demodulated DC value and not a transient response.

3.4. Overview

To verify the operation of the Lock-In Amplifier (LIA), two simulation cases were performed in LTspice. The reference frequency of the LIA was set to **1 kHz**. An input signal was applied at two different frequencies to confirm that the LIA correctly extracts only the signal at the reference frequency and rejects all others.

3.5. Case 1: Input Signal at 1 kHz (On-Frequency)

A sinusoidal input signal of amplitude 10 mV at **1 kHz** was applied, matching the LIA reference frequency. As expected, the AD630 synchronous demodulator produces a rectified signal, which after passing through the second-order active low-pass filter yields a stable **non-zero DC output voltage**.

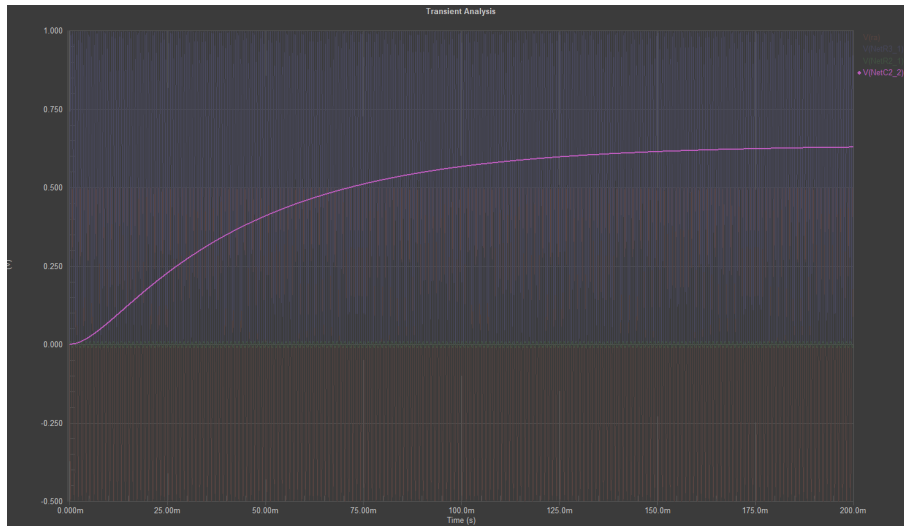


Figure 4: Simulation output at 1 kHz input — DC output present at the filter output, confirming signal detection at the reference frequency.

3.6. Case 2: Input Signal at 2 kHz (Off-Frequency)

The input frequency was changed to **2 kHz**, which is twice the reference frequency and thus out of lock. The synchronous demodulation process now produces an AC signal at 2 kHz with zero mean. The low-pass filter completely attenuates this, resulting in **effectively zero DC output**, confirming the frequency selectivity of the LIA.

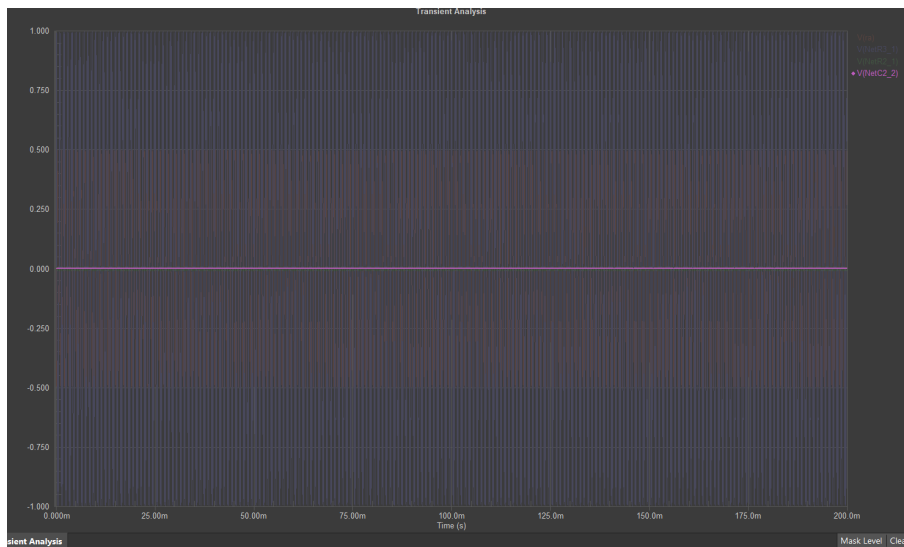


Figure 5: Simulation output at 2 kHz input — DC output collapses to zero, confirming rejection of off-frequency signals.

3.7. Conclusion

The simulation results confirm the correct operation of the designed Lock-In Amplifier. A clear DC output was obtained when the input signal matched the 1 kHz reference frequency, while the output collapsed to zero when a 2 kHz off-frequency signal was applied. This demonstrates that the LIA successfully performs synchronous demodulation with strong rejection of out-of-band signals, validating the design.

4. Conclusion

This report has described the design of a dual-channel IQ lock-in amplifier based on the AD630. The key design features are:

1. A **preamplifier** with gain of 50 using the ADA4610 to boost weak input signals
2. A **phase-sensitive detector** using the AD630 with gain of 2
3. A **frequency-independent IQ reference generator** using the CD4046BE PLL and SN74HC74 flip-flops, providing exact 90° quadrature at any frequency
4. A **10 Hz low-pass filter** to extract the DC component from both I and Q channels
5. **Amplitude and phase recovery** using:

$$A = \frac{\pi}{200} \sqrt{X^2 + Y^2} \quad (20)$$

$$\phi = \arctan\left(\frac{Y}{X}\right) \quad (21)$$